DIY-AI

Running Large Language and Agentic Models in your own environments

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Local AI: models, runtimes and hardware

A trained **model** supplies the weights. A **runtime** loads them and runs inference on compatible **hardware**.

Runtime	Example model	Workshop use
Ollama service	`gemma4:26b`	Text chat on Mac
vLLM Metal service	`mlx-community/gemma-4-e4b-it-4bit`	Text, no audio in this setup
vLLM Metal service	`openai/whisper-large-v3`	Audio transcription
MLX-VLM program	`google/gemma-4-E4B-it`	Audio input on Mac
vLLM service	`deepseek-ai/DeepSeek-V4-Flash`	Text and code on Spark

Local AI is practical: match your task and model to the runtime and available memory.

Workshop configurations from the outline, notebooks and launch scripts. Runtime references: vLLM Metal, MLX-VLM, Ollama.

Four ways to run the workshop models

Hardware and runtime	How we use it	Entry point
Apple Silicon + Ollama	Easiest starting point for local chat	`ollama run gemma4:26b`
Apple Silicon + vLLM Metal	A local model service for notebooks and apps	`./scripts/start-mlx-vllm.sh` API at `localhost:8000`
Apple Silicon + MLX-VLM	Direct Python inference, including Gemma audio	`notebooks/local-vlm-gemma.ipynb`
DGX Spark + vLLM	Connect over VPN and call the remote model service	Change the API base URL to Spark

After installation and download, start a model with a command or a cell.
Gemma's saved local model load: **4.8 seconds**.

Sources: local launch scripts and notebooks. Spark audio uses 1 DGX Spark. DeepSeek uses 2 DGX Sparks. Hardware and VPN configuration follow the workshop outline.

Three tests of useful model behavior

1. Five questions: responsiveness and completeness

Capital of France, an AI poem, quantum computing, the laws of robotics and the plot of *Inception*.

Measures: request latency, completion tokens per second and cut-off answers.

2. Dutch transcription: agreement with a curated reference

Jeroen Hendrikse's radiology recording from UMC Utrecht.

Measure: word error rate (WER) over 1,350 reference words.

3. LiveCodeBench-Pro: executable programming solutions

Ten contest problems, with one C++ solution attempt per problem.

Measure: accepted solutions out of ten selected problems.

Only DeepSeek on Spark produced judgeable code among our self-hosted setups.

GPT-6 Astra provides a separate cloud comparison.

Sources: original presentation, transcription notebooks and saved LiveCodeBench-Pro results. The five-question test is a smoke test, without a systematic correctness score.

Five questions: speed and completion

Request latency	**Local vLLM** Gemma E4B	**Ollama** Gemma 26B	**Spark** DeepSeek V4 Flash
Capital of France	1.745 s	1.478 s	**0.607 s**
AI poem	5.491 s	19.619 s	**3.686 s**
Quantum computing	7.818 s	22.523 s	23.756 s
Laws of robotics	7.463 s	14.958 s	8.349 s
Inception	7.794 s	24.565 s	15.920 s
Mean latency	6.062 s	16.628 s	10.464 s
Completion rate (tokens/s)	18.64	34.37	28.94
Incomplete answers	3 of 5	2 of 5	0 of 5

Local vLLM's three faster answers were cut short.

Spark finished all five responses with `stop`.

Reused from the original deck's 7 October 2026 snapshot. Token budgets: 150 / 800 / 800. Different models, tokenizers and reasoning limit comparisons. Completion rate = total completion tokens / total request time.

Dutch transcription: three models on one recording

Jeroen Hendrikse, UMC Utrecht: *radiologieverslag*
473.6 seconds of speech, scored against 1,350 curated reference words.

Model	**Runtime / hardware**	**WER ↓**	**Time**
Whisper large-v3 `openai/whisper-large-v3`	vLLM Metal Apple Silicon	12.00%	Not recorded
Gemma 4 E4B `google/gemma-4-E4B-it`	MLX-VLM Apple Silicon	**7.78%**	144.3 s
Qwen3-Omni 30B A3B `Qwen/Qwen3-Omni-30B-A3B-Instruct`	vLLM 1 DGX Spark	8.44%	98.1 s

Gemma has the lowest WER. Qwen finishes the transcription loop sooner.

Sources: saved outputs of notebooks/local-llm-whisper.ipynb, notebooks/local-vlm-gemma.ipynb and notebooks/local-llm-qwen-omni.ipynb. Time covers the transcription loop, excluding model loading and initial audio decoding. Whisper's notebook does not time the request.

Word error rate: Gemma's small advantage

$$\mathrm{WER} = \frac{\text{substitutions} + \text{deletions} + \text{insertions}}{\text{reference words}} \times 100\%$$

Model	Substitutions	Deletions	Insertions	Total
Whisper large-v3	86	41	35	162
Gemma 4 E4B	69	21	15	**105**
Qwen3-Omni	73	15	26	114

Gemma: 9 fewer word errors than Qwen

7.78% versus 8.44% WER: a **0.67 percentage-point** gap using the unrounded scores.

Scoring ignores case and punctuation, splits hyphens, and retains fillers and repetitions. Curated TXT and SRT references yield identical scores.

Sources: the three transcription notebooks, cell index 7. One recording and different inference pipelines. This small difference does not establish a general model ranking. WER counts word differences without weighting their clinical meaning.

LiveCodeBench-Pro: a harder programming test

Contest problems from **Codeforces, ICPC and IOI** require an algorithm and working code that passes the judge's tests.

What happened locally

Local MLX trials did not yield usable solutions in the workshop.
The saved comparison records:

- **Ollama, Gemma 26B:** all 10 responses hit the 4,096-token limit without final code.
- **Local vLLM: Not run.**

The larger model

DeepSeek V4 Flash

`deepseek-ai/DeepSeek-V4-Flash`

vLLM over VPN on **2 DGX Sparks**
256 GB total unified memory
(128 GB per node)

One C++ attempt per problem.
Temperature 0, 4,096 tokens.

The trials expose task difficulty and a token-budget limit, without establishing a general limit for local models.

Local MLX observation and cluster configuration: workshop outline. Saved evidence: Ollama, 8 October; Spark, 20 October 2026. LiveCodeBench-Pro paper; DGX Spark specifications.

LiveCodeBench-Pro: Spark and GPT-6 Astra

Same ten problems and ARM64 judge. Both runs completed judging for all ten.

	DeepSeek V4 Flash 2 DGX Sparks, 20 October	**GPT-6 Astra** Codex session, 21 October
Accepted / selected	**4/10 (40%)**	**10/10 (100%)**
Compilation error	4	0
Wrong answer	2	0
Mean inference time	25.32 s	Not measured

Astra passed all ten problems. DeepSeek's six failures included four compilation errors.

Generation conditions differ:

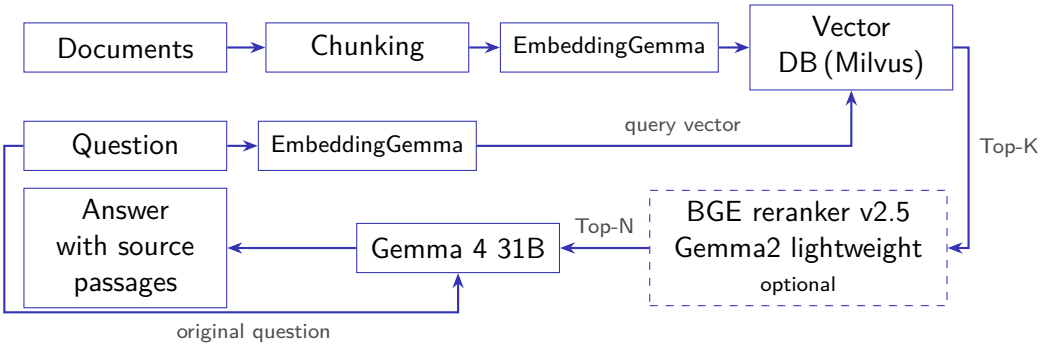
DeepSeek: temperature 0, 4,096-token API budget per problem.

Astra: Codex with `xhigh` reasoning, shared conversation context, no per-problem token cap.

Source: livecodebench-gpt-6-astra-results.tex and saved JSON. One judged submission per problem. Astra sources frozen before testing, with no revisions after feedback. Subset: 7 easy, 3 medium, 0 hard. Generation conditions differ. These are workshop results, not full-benchmark leaderboard scores.

Local document search with RAG on DGX Spark

Retrieval-augmented generation (RAG) supplies relevant document passages to the model before it writes an answer.



Top-K retrieves candidate passages. The optional reranker selects Top-N. Without reranking, the retrieved passages go directly to Gemma.

Proposed local stack from the outline, not a benchmarked deployment. Model references: EmbeddingGemma 300M, BGE reranker, Gemma 4 31B; Milvus documentation.

Embeddings: meaning as coordinates

Text becomes a vector

An embedding is a **list of numbers** representing text.

Similar meanings tend to lie nearby in a high-dimensional space.

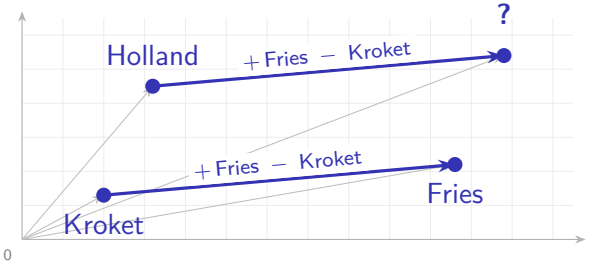
Meaning has direction

$$E(\text{King}) + E(\text{Woman}) - E(\text{Man}) \approx E(\text{Queen})$$

$E(\text{word})$ is its embedding vector.

Quiz: which country?

Holland + Fries − Kroket ≈ **?**



Illustrative 2D embedding space; not a measured model result.

Back to RAG: EmbeddingGemma represents questions and document chunks in a shared space so similarity search can retrieve relevant passages.

Embedding intuition and word analogy: 3Blue1Brown, Transformers, the tech behind LLMs. Text retrieval: EmbeddingGemma model card.

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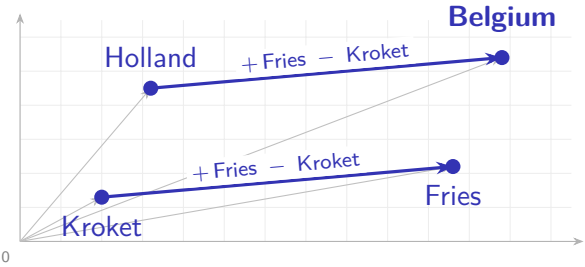
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Holland + Fries − Kroket ≈ **Belgium**



Illustrative 2D embedding space; not a measured model result.

Add the fries association and subtract the kroket association.

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Next: agentic orchestration in clinical imaging

Local models and document retrieval are building blocks for a wider workflow.

The next presenter will introduce an agentic orchestrator for use in clinical imaging practice.

Coordinating model calls, retrieved information and tools around the steps of a clinical imaging workflow.

Over to the next presenter.